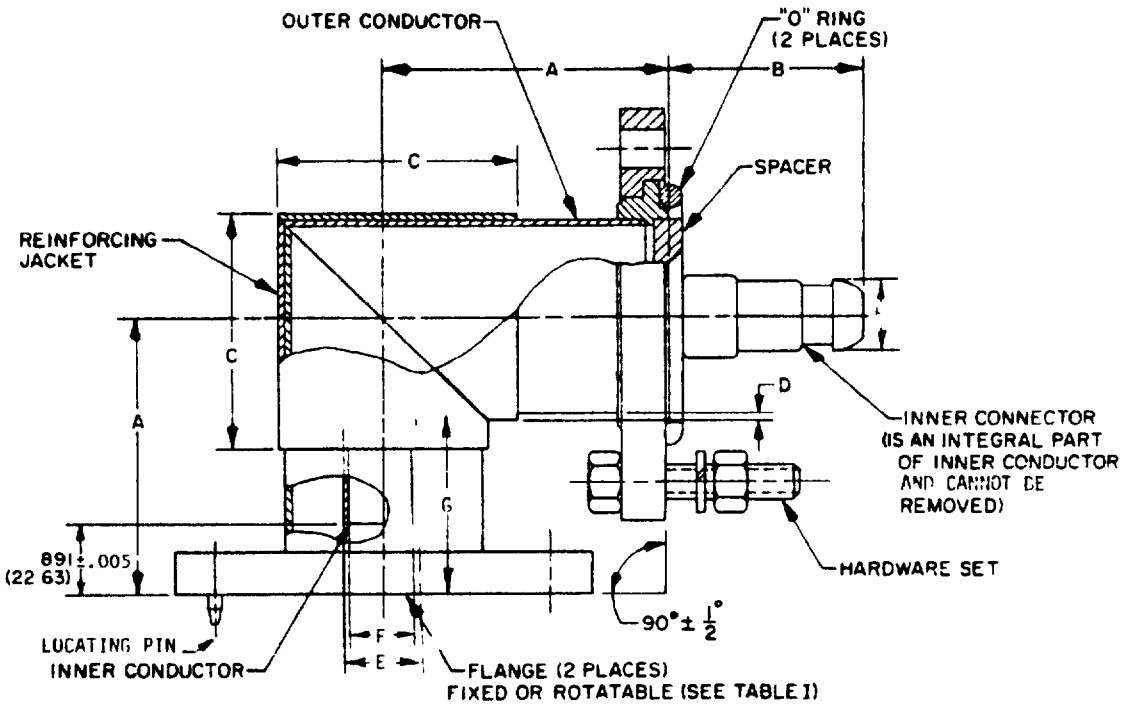
MILITARY SPECIFICATION

LINE ASSEMBLIES, RADIO FREQUENCY TRANSMISSION
3-1/8 INCH ELBOWS

The complete requirements for procuring the elbow assemblies described herein shall consist of this document and the latest issue of Specification MIL-L-28796.

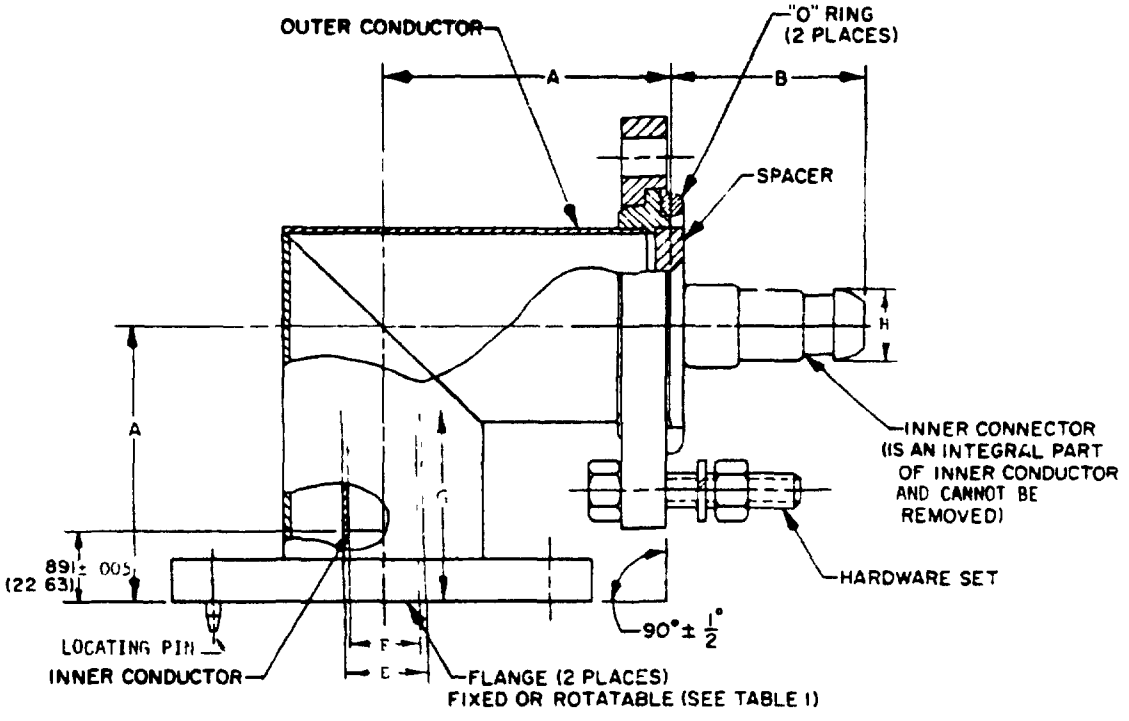


NOTES
1 Dimensions are in inches
2 Metric equivalents are in parentheses
3 Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm

FIGURE 1 90 degree miter copper elbow assemblies

FSC 5985

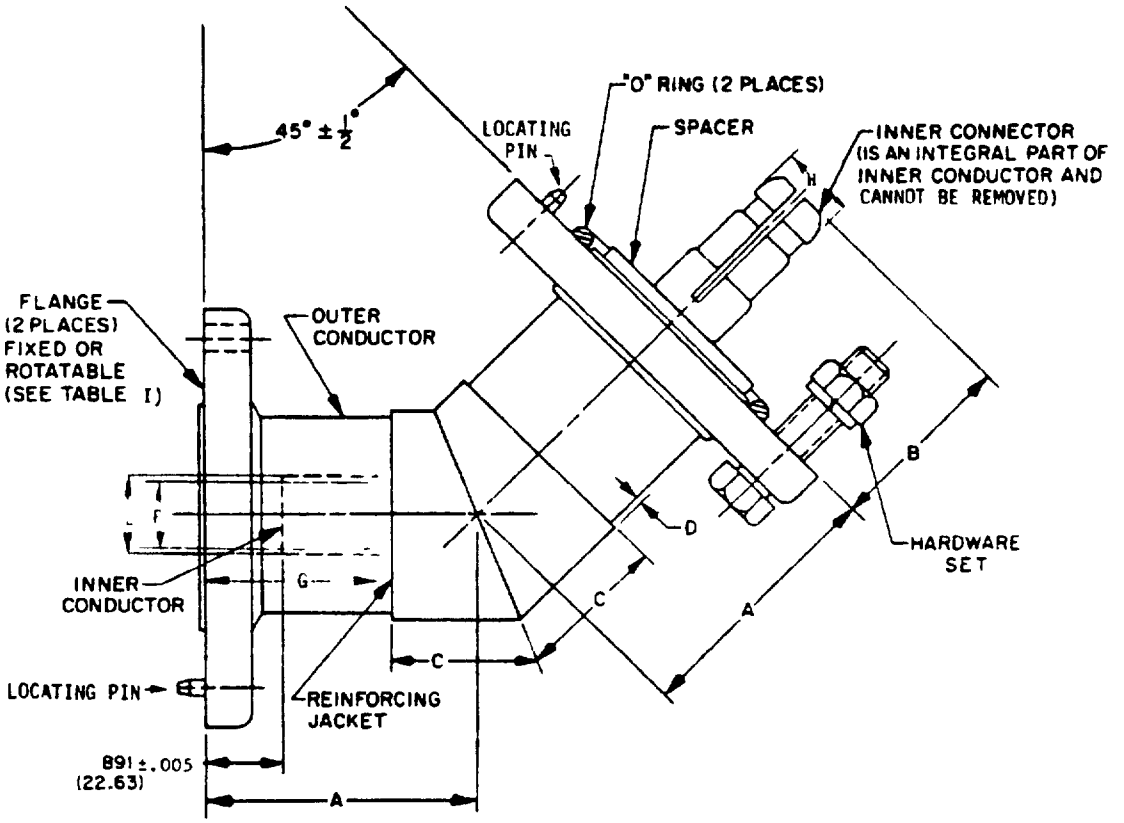
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NOTES
1 Dimensions are in inches
2 Metric equivalents are in parentheses
3 Metric equivalents (to the nearest 01 mm) are given for general information only and are based upon 1 inch = 25 4 mm

FIGURE 2 90 degree miter aluminum elbow assemblies

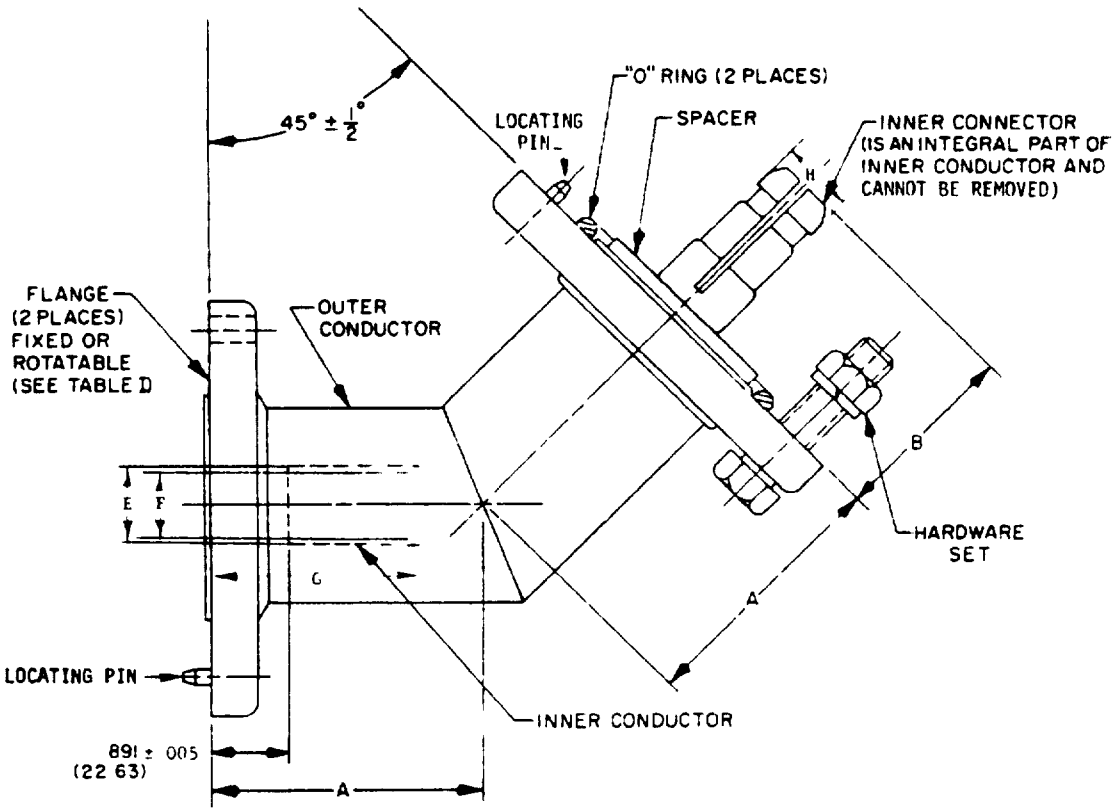
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NOTES
1 Dimensions are in inches
2 Metric equivalents are in parentheses
3 Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm

FIGURE 3 45 degree miter copper elbow assemblies

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NOTES
1 Dimensions are in inches
2 Metric equivalents are in parentheses
3 Metric equivalents (to the nearest 01 mm) are given for general information only, and are based upon 1 inch = 25 4 mm

FIGURE 4 45 degree miter aluminum elbow assemblies

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REQUIREMENTS

Design and construction

Dimensions and configuration See figures 1 through 4 and table I

Operating frequency 400 to 450 MHz

Gaskets Class 2b, grade 50 of ZZ-R-765.

Hardware set Six each, corrosion - resistant steel bolts, nuts, and lockwashers, 3/5-16 UNC X 1-1/2 inches long

Finish
Inner conductor - Silver plate per QQ-S-365, type I
Outer conductor and flanges (aluminum) - Chemical film per MIL-C-5541 type II, grade C, class 3

Surface roughness 63 microinches

Insertion loss .05 dB maximum.

VSWR 1.02 1 maximum.

Impedance 50 ohms ±.3 ohms.

Velocity of propagation 99 7 percent minimum

RF Power handling capability
Average - 40 kilowatts minimum.
Peak - 390 kilowatts minimum.

Dielectric withstanding voltage· 20 kilovolts (ac)

Pressurization·
Test - 65 lbf in^2G $^{+2}_{-0}$ lbf in^2 G.

Operating - 20 lbf/in^2G ±2 lbf/in^2 G.

Part number M28796/2 -(and dash number from table I)

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TABLE I Dash numbers and characteristics

Dash number	Flange per MIL-F-24044/	Transmission line per MIL-L-3890/ type	Inner connector per MIL-F-24044/	Dimensions			
				A ±.016	B ±.016	C ±.031	D +.010 -.000
-001	1-09 (fixed)	1-RG-154/U	1-09C	3.812	2.125	3.412	.125
-002	2-09 (rotatable)	1-RG-154/U	2-09C	3.812	2.125	-	-
-003	1-10 (fixed)	1-RG-154/U 1/	1-09C	3.812	2.125	3.412	.125
-004	2-10 (rotatable)	1-RG-154/U 1/	2-09C	3.812	2.125	-	-
-005	1-09 (fixed)	1-RG-154/U	1-09C	3.812	2.125	3.412	.125
-006	2-09 (rotatable)	1-RG-154/U	2-09C	3.812	2.125	-	-
-007	1-10 (fixed)	1-RG-154/U 1/	1-0	3.812	2.125	3.412	.125
-008	2-10 (rotatable)	1-RG-154/U 1/	2-09C	3.812	2.125	-	-

Table I Dash numbers and characteristics (Cont'd)

Dash number	Dimensions				Figure number
	E ± 003	F ± 003	G 2/ Min	H + 000 - 002	
-001	1.315	1.231	2.141	1.240	1
-002	1.315	1.231	2.141	1.240	1
-003	1.315	1.231	2.141	1.240	2
-004	1.315	1.231	2.141	1.240	2
-005	1.315	1.231	2.141	1.240	3
-006	1.315	1.231	2.141	1.240	3
-007	1.315	1.231	2.141	1.240	4
-008	1.315	1.231	2.141	1.240	4

1/ The same as RG-154/U, except the material shall be aluminum (6061)
2/ Distance which straight section of inner conductor with specified inside and outside diameter shall extend to allow insertion of inner connector

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Review activities:

Navy - SH, OS, AS

User activities

Navy - MC

Preparing activity:
Navy - EC

Agent.
DSA - ES

(Project 5985-N393-2)

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OMB Approval No 22-R255

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